VF Asia Footwear Laboratory Testing Manual

Test Method for Hardware Attachment and Lace Holder Strength	Test method # VF- B04 REF #: TM-R- 016 Page 1 of 4
Date Issued: May, 2019 Revision #: 4 Reason for Revision: Rectify the unit of measure from kg to kgf for strength of material.	Created by: Nila Babia Erazo

PURPOSE AND SCOPE: To determine the strength of any type of footwear fastening that involves the use of a lace when it is subjected to a force applied by the lace in the same manner as in wear. Additionally, in the case of eyelets, this method is used to assess the security of the eyelet clenching by measuring the load required to force the eyelet out when applied in a direction perpendicular to the facing.

REFERENCES: SATRA TM 149, SATRA TM 150

EQUIPMENT AND MATERIAL:

1. Tensile testing Machine with clamping jaws, preferably rubber-faced or with pneumatic grips.
2. A flat – woven nylon or polyester shoe lace with a minimum tensile breaking strength of 50 kgf
3. Strong upper leather, for use when test pieces need to be prepared.
4. Eyelet testing jig
5. Hardware such as eyelet, Ski hook, D-ring

SAMPLE PREPARATION:

A. Strength of Eyelet Facings, Ski Hooks and D Rings

TEST SAMPLE of RAW MATERIAL:

1. Die or hand cut a sample of the material which is 30 ± 1mm in width by at least 60mm in length.
2. Punch a 3.0+/-0.1mm hole in the test sample, such that the center of the hole is located 9.5 ± 0.5mm from the short edge of the test sample and centrally located between the long edges.

See Figure 1

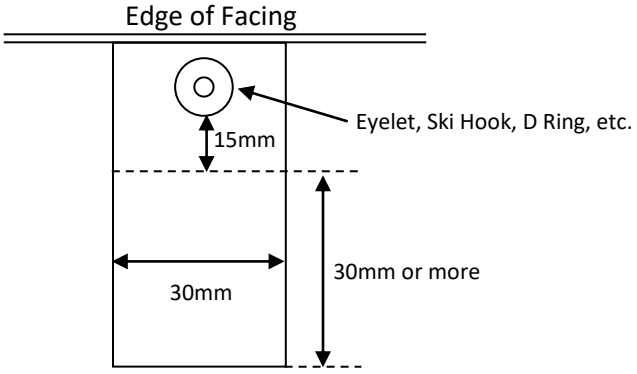


Figure 1. Standard test piece for testing an eyelet.

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TEST SAMPLE from ACTUAL UPPER:

1. Remove the outsole from upper.
2. Separate the test sample extracted from lateral side and from medial side. This is to avoid the upper touching the tensile machine during testing.

B. Security of Eyelet Attachment in Footwear:

Measure to the nearest 0.5mm the internal diameter of the eyelet hole and the external diameters of the eyelet flange and clenching.

C. Lace Holder:

1. The test sample of lace holder should be taken from the actual upper material. If the sample submitted is the whole footwear, the outsole should be removed.

REPETITONS: Test three (3) samples or if sample not enough as per availability of sample

TEST PROCEDURE:

Testing should be conducted at 23 ± 2 ℃ Temperature and 55 ± 10% relative humidity.

A. *Strength of Eyelet Facings, Ski Hooks and D Rings:*

TEST SAMPLE of RAW MATERIAL

1. Set the rate of traverse of the tensile tester to 100 mm/min.
2. Clamp the test piece centrally in the tensile tester such that the edges of the jaws align along the dashed line in Figure 1.
3. Cut a length of lace at least 200mm long. Insert the lace through the fastening, equalize the ends, and clamp them into the second jaw.
4. Operate the tensile testing machine and observe the fastening. As soon as any failure occurs, record the corresponding force. Continue the test until complete failure occurs. Record the maximum force to cause failure.

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TEST SAMPLE from ACTUAL UPPER:

1. Insert the shoe lace into the eyelet.
2. Clamp the shoe lace at one jaw and the upper (eye-stay) at the other jaw. ***(Note: If the distance from the edge of the eyelet to eye-stay is too narrow that clamping is difficult, clamping will be made on the upper. To avoid premature breakage of upper, reinforce with stronger upper material)***
3. Set the speed at 100 mm/min. Operate the tensile testing machine. As soon as any failure occurs, record the corresponding force. Record the maximum force to cause failure.

B. Security of Eyelet Attachment in Footwear:

1. Fit the compression cage of the testing jig into the tensile tester so that it is exactly vertical and the conical finger points downwards.
2. Balance the load cell so that it reads 0 kgf.
3. Set the rate of traverse to 100 mm ± 20 mm/min.
4. Select the circular clamp that is appropriate to the size of the eyelet. For eyelets whose larger diameter is less than 10mm use the clamp with a 12.5mm central hole, but for eyelets whose larger diameter is 10mm or more, use the clamp with the 16mm hole.
5. With the screw heads of the clamp downwards lift off the top plate. Position a test piece on the bottom of the plate so that the eyelet is central relative to this plate, with the clenched side upwards for standard eyelets and downward for blind eyelets. Replace the top plate of the clamp and tighten it firmly and evenly.
6. Position the circular clamp with the test piece in it centrally on the lower platform of the cage with the screw heads downwards.
7. Operate the tensile tester so that the tip of the conical finger first enters the eyelet and the sides of the finger then make contact.
8. Measure and record the maximum force required to force the eyelet out of its facing.

***C.* Lace Holder:**

1. Insert the shoe lace into the lace holder.
2. Clamp the lace holder at one jaw and the shoe lace at the other jaw. Ensure that the samples are aligned.
3. Set the speed at 100 mm/min. Operate the tensile testing machine and observe the fastening. As soon as any failure occurs, record the corresponding force. Continue the test until complete failure occurs. Record the maximum force to cause failure.

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REPORT:

A. Strength of Eyelet Facings, Ski Hooks and D Rings

- Maximum force at failure in kgf
- Type of failure
- Any deviations from this procedure.

B. Security of Eyelet Attachment in Footwear:

- Maximum force at failure for detachment in kgf
- Internal diameter of the eyelet hole
- Diameter of the flange,
- Diameter of the clenching,
- Any deviations from this procedure.

C. Lace Holder:

- Maximum force at failure in kgf
- Type of failure
- Any deviations from this procedure.